

badly. With one hundred assets, there are 5,510 parameters to estimate.³³ For five thousand stocks (approximately the number of common stocks listed in U.S. markets) the number of parameters to estimate exceeds 12,000. The potential for errors is enormous.

Let's view what happens when we move from the optimal mean-variance strategy and turn off some of the inputs, so that we are relieved from estimating means, volatilities, correlations, or combinations of all three. Table 3.16 lists some special cases as the restrictions are imposed.

The minimum variance portfolio is a special case of full mean-variance that does not estimate means and in fact assumes that the means are all equal. Risk parity is a special case of mean-variance that does not estimate means or correlations; it implicitly assumes that all assets have the same mean and all assets are uncorrelated. The equally weighted portfolio has nothing to estimate. It is also a special case of mean-variance and assumes all assets are identical.

I have included market weights in Table 3.16. Like equal weights, there are no parameters to estimate using market weights. The important difference between equal weights and market weights is that the equal weighted portfolio is an *active strategy*. It requires trading every period to rebalance back to equal weights.

Table 3.16

	<i>Assumptions on Means</i>	<i>Assumptions on Volatilities</i>	<i>Assumptions on Correlations</i>	<i>Comments</i>
Optimal Mean-Variance	Unconstrained	Unconstrained	Unconstrained	Most complex
Minimum Variance	Equal	Unconstrained	Unconstrained	No need to estimate means
Risk Parity	Equal	Unconstrained	Equal to zero	No need to estimate means or correlations
Equally Weighted (1/N Portfolio)	Equal	Equal	Equal	Most simple and active, Nothing to estimate
Market Weight	–	–	–	Observable and passive, Nothing to estimate

³³ For N assets, you have N means and $N \times (N + 1)/2$ elements in the covariance matrix.